Doc Link:-
https://docs.google.com/document/d/1ZNcs3f8CnHKmdMIh9siDrp8uO57CnV0sJjiHKgQ7w78/edit?usp=sharing

Chapter link:- https://jax-ml.github.io/scaling-book/applied-inference/

Model	Pod size	Host size	HBM capacity/chip	HBM BW/chip (bytes/s)	FLOPs/s/chip (bf16)	FLOPs/s/chip (int8)
TPU v3	32x32	4x2	32GB	9.0e11	1.4e14	1.4e14
TPU v4p	16x16x16	2x2x1	32GB	1.2e12	2.75e14	2.75e14
TPU v5p	16x20x28	2x2x1	96GB	2.8e12	4.59e14	9.18e14
TPU v5e	16x16	4x2	16GB	8.1e11	1.97e14	3.94e14
TPU v6e	16x16	4x2	32GB	1.6e12	9.20e14	1.84e15

Host size refers to the topology of TPUs connected to a single host (e.g. TPU v5e has a single CPU host connected to 8 TPUs in a 4x2 topology). Here are interconnect figures:

Model	ICI BW/link (one-way, bytes/s)	ICI BW/link (bidi, bytes/s)
TPU v3	1.0e11	2.0e11
TPU v4p	4.5e10	9.0e10
TPU v5p	9.0e10	1.8e11
TPU v5e	4.5e10	9.0e10
TPU v6e	9.0e10	1.8e11

Let's remind ourselves what LLaMA 3-70B looks like (see Section 6 for reference):

hyperparam	value
(L)	80
(D)	8,192
(F)	28,672

(N)	64
(K)	8
(H)	128
(V)	128,256

Let's start with a simple question: what hardware should we serve on? The answer is basically, whichever is cheapest in FLOPs / dollar.[1] For this reason, we typically want to serve on TPU v5e, our current dedicated inference chip (cost comes from Google Cloud pricing as of February 2025):

TPU type	bfloat16 FLOPs/s	Google Cloud USD / hour	FLOPs / $
H100	9.9e14	$10.8	3.3e17
v5p	4.59e14	$4.2	3.9e17
v5e	1.97e14	$1.2	5.8e17

Each TPU v5e has 16GB of HBM which will require us to shard our model fairly aggressively.

Question: How large are LLaMA 3-70B's KV caches per token? *You can assume we store them in int8. This determines how large our batch size can be on a given topology.*

My Answer:-
per token KV bytes=2KHL=164KB

Question: Let's say we want to serve L3 70B at batch size 32 and 8192 sequence length with everything (params and KVs) in int8. How much total memory will this use? What's the smallest slice we could serve this on?

My Answer:-
model size=70GB
total KV =2*KHLBS=43GB

total memory req=70+43=113GB

memory per chip=16GB so total chips req>=7

so 4x4 TPU topology works

Question: At this batch size and quantization on a TPU v5e `4x2`, roughly what latency would we expect per decode step? What throughput (tokens / sec / chip). What about a `4x4`? *Assume we perform our FLOPs in bfloat16 and everything is fully sharded.*

My Answer:-
our toppology is 4X2
time required to load KV caches=43gb/8*Whbm
Whbm=43e9/16*8.1e11=6.6ms

not for MLP time req=Max(time to lead model from HBM to MXU, compute time)
=max(parameter count/8Whbm, 2*batchSize*ModelParams/Flops)
=max(70e9/8*8.1e11,2*32*70e9/8*1.97e14)
=max(11ms,3ms)
=11ms

so total time=17 ms for 4x2 topology and 32*1000/17=235 tokens/chip/sec

for 4x4 it will be half so it is 8.5ms

for 2*batchSize*ModelParams/Flopsto be valid we have to see when it is cm[pute or commune bound in ici and MXU for MLP layer so we have compute bound when F>YC/Wiciso its 4*1.97e14/9e10=8755

yeah check out it is compute bound

Question: On TPU v5e, using bfloat16 weights and activations, how large do our batch sizes need to be for us to be compute-bound in our matmuls? What if we do int8 weights but perform our FLOPs in bfloat16? What about int8 weights with int8 FLOPs?

My Answer:-
to be compute bound on MLP layer we need
Tcompute>Thbm
so 2*batch_size*param_count/C>model_size/Whbm
batch_size>C/2Whbm
for FLOPs anf b16 byte Whbm is half
batch size>1.97e14/8.1e11
Batch_size>240

for FLOPs and int8
it is B>120

for int ops and int8
B>240

Question: What is the smallest TPU v5e topology we could serve LLaMA 3-70B on using bfloat16, int8, and int4 (both KVs and parameters) with 8k context? *You can think of KV caches as negligibly small for this one.*
My Answer:-

considering memory only as bottleneck constrain
Tokens per batch is 8e3
so we should have total memory>=param_size*(bytes per param)

so we only have to fit in model for this one
so 70GB*2/16=nearly 4x4 is prefered for bf16
for int8 it can be 4x2 and for int4 its 2x2
but to use larger batch size B>240 we preffer larger topology

Question: Assume we use the largest batch size that fits on these topologies, what latency could we expect for each generate step?
My Answer:-
with 8k in context we have out KV cache size per batch at around of
2SKHL=2*8e3*8*128*80=2.6GB per batch for bf16
so around 43 batch size is possible

now time required is aroud for per chip of 16GB/Whbm for per chip generation bnecause it is majorly commute bound because of low batch size
which is around 16e9/8.1e11=20 ms per chip so like per chip per sec around 50 tokens
so for 16 chips its around 800 tokens this is for int 8 for bf16 its 40ms per token and 10 ms for int4

Question: For each of these, what throughput per chip does this give us (in terms of queries / chip)? *You can assume our median decode length is 512 tokens.*

My answer:-
for int8 we have 43 kv caches batch size and 16 chips and 20 ms per chip per token so tota for 4x4 topologyl tokens generated total=1000/20=50 steps per sec so total =50*43/16/512=0.26 qps for dsame topology int8 and int4 will generate 0.5 qps per chip and 1 qps per chip

Question: How would our peak throughput change if we doubled our topology for each of the above examples?
My answer:-
for bf16 we will count number of KV means batch size we can give

16*16*2=512GB so available for kV=512-70*2=372GB
per KV memory batch =2.6GB
so total=143KV
so we will have qps multiplied by 143/43=3.3 times thropughput total
but for per chgip it will be 3.3/2=1.6 times around of previous results

Question: Now let's dig into the question of sharding. Let's say we wanted to serve in bfloat16 on a TPU v5e 4x8. What sharding would we use for our model on a TPU v5e 4x8 during generation? Can we avoid being communication bound?

My Answer:-
We have 143 batch size so we are
its test for the tensor how much is for compute bound we need F>32/2*2200=35000 so it is ici commune bound when using tensor parrelelism
now lets see it is Thbm bound or Tici bound
so we need Y>F/(B* beta)
so Y>28672/(143*8)=25 and we have 32 TPU chips so it is Tici bound
so we better serv on 4x4 TPU because in model parrelelism its ici communication bound
or if we do communication in int8 params and compute in bf16 params then we will be copmpute bound

Question: Assume we achieve a 40% FLOPs utilization during prefill. How long will a prefill of length 8192 take on 16 TPU v5e chips?

My Answer:-
because we can do FSDP or tesonr parrelelism for full use
time per step= (1/e)(2*model_params*8192)/C=2*8192*70e9/(0.4*16*1.97e14)=0.9s

Question: Assume we have a median prefill length of 8192 tokens and a median decode length of 4096 tokens. Say we have a generate batch size of 32. On average how many sequences finish decoding per step? On average how many tokens are evicted from our KV cache each step?

My Answer:-
for perfill we take 0.9s

now we will find time per step of decoding it will be 70GB for model loading and 2*KHLBS
where S is on avg 4096+8192=12288
so total KV memory in params =64.4GB
so per step considering int8 time=70+64.4e9/8.1e11=0.16s per step
so per token time is like 0.16/32=0.5ms
so per step on avbg 32/4096 seq finish decoding seq finishes

avg kv cache 32*(8192+4096)/4096

Question: Assume we do disaggregated serving with a median prefill length of 8192 and a median decode length of 512. Assume the prefill and generate latencies calculated above in bfloat16. What ratio of prefill:generate servers will you need to keep both fully saturated.

My Answer:-

we need that prefil and generate takes similar time
now prefill num seconds= 0.9/Nprf for batch of size 1
for generate it for 32 batch it takes around 0.16/Ngen secs per step so so like Ngen/0.16*32 tokens per step
for 32 batch the memory is 70GB+2KHLBS
S=8192+512=8704
so KV size=45.6GB
so time taken for 32 batch is 2*(70+45.6)/16*8.1e11=285ms/16 per step of 32 tokens 2 multiplied because of bf16
so total for 32 seq to ent it takes 0.142*512/Ngen steps

so 0.9*32/Nprf=0.285*512/16*Ngen
so Nprf=3Ngen